

The Converse for Spherically Symmetric Trees

A candidate full solution to the question after Theorem 2.2 of arXiv:1008.3418

Automated mathematics run `fa_banach_001`, lane 9

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Status. Candidate full solution, likely valid, pending expert review. The argument is self-contained apart from the standard equivalence between generalized roundness and supremal negative type recalled in the source paper.

1 Source question and answer

Caffarelli, Doust, and Weston prove in Theorem 2.2 of [1] that if the downward degree sequence $(d_0, d_1, \dots)$ of a countable spherically symmetric tree (SST) has infinitely many entries larger than one, then the tree has generalized roundness one. On page 5, immediately before Section 3, they ask:

“We do not know if the converse of Theorem 2.2 is true.”

The answer is affirmative. We prove a stronger metric-tree statement. A *finite-topological tree* will mean a geodesic metric tree obtained by gluing finitely many compact intervals and half-lines at endpoints, without creating a cycle. Its open edges are the interval or half-line interiors after a finite vertex set has been chosen. Subdivision only increases the number of open edges, so no minimality convention is needed.

2 Proof intuition

For a finite signed configuration of total mass zero on a tree, there is a unique cumulative flow along each oriented edge. Integrating the distance-power form twice transfers the coefficients to these flows. On one edge the mixed derivative of $|s - t|^p$ has the favorable sign and produces the positive Riesz kernel $|s - t|^{p-2}$. Between two distinct edges, the distance is affine in the two edge coordinates, so the unfavorable contribution has the shifted Hankel kernel $(s + t + c)^{p-2}$.

The key point is quantitative: when p is just above one, every Hankel cross term is a small fraction of the corresponding within-edge Riesz energies. A logarithmic change of variables converts both homogeneous kernels into convolution kernels. Their Fourier-multiplier ratio is exactly

$$\frac{\cos(\pi\alpha/2)}{\cosh(\pi\xi)}, \quad \alpha = 2 - p.$$

Hence the relative norm is $\cos(\pi\alpha/2) = \sin(\pi(p-1)/2)$, which tends to zero as $p \downarrow 1$. There are only finitely many edge pairs, so diagonal dominance proves p -negative type for some $p > 1$.

Theorem 2.2. *Let (T, ρ) be a countable SST with degree sequence $(d_0, d_1, d_2, \dots)$. If $|\{j \mid d_j > 1\}| = \aleph_0$, then (T, ρ) has generalized roundness one.*

Proof. Let (T, ρ) be a given countable SST as in the statement of the theorem. The condition on $(d_0, d_1, d_2, \dots)$ ensures that $d_1 d_2 \cdots d_k \rightarrow \infty$ as $k \rightarrow \infty$. Moreover, by considering the truncations of T , it is clear that $\gamma(T)$ satisfies (4) for all $n \geq 3$ and all $k < n/2$. Now set $k = \lfloor \ln n \rfloor$ (for example) and let $n \rightarrow \infty$. We see immediately that $\gamma(T) \leq 1$. But, as we noted at the outset of this section, $\gamma(T) \geq 1$ too. Thus, $\gamma(T) = 1$. $\square$

One immediate consequence of Theorem 2.2 is the following observation.

Corollary 2.3. *There exists an uncountable number of mutually non isomorphic countable SSTs of generalized roundness one.*

Proof. Consider the set of degree sequences

$$D = \{(1, d_1, d_2, d_3, \dots) : 1 < d_1 < d_2 < d_3 < \dots\}.$$

This set is clearly uncountable, and by Theorem 2.2 the SST corresponding to each element of D has generalized roundness one. Suppose that $(1, d_1, d_2, \dots)$ and $(1, d'_1, d'_2, \dots)$ are distinct elements of D , and let T and T' be the corresponding SSTs. Let k be the smallest integer such that $d_k \neq d'_k$ where, without loss, we may assume that $d_k < d'_k$. Then T has vertices with degree $d_k + 1$, but T' does not, and so the two graphs are not isomorphic. $\square$

Virtually all countable SSTs encountered in analysis, topology and discrete geometry satisfy the condition placed on the downward degree sequence in the statement of Theorem 2.2. This is the case for k -regular trees ($k \geq 3$), complete n -ary trees of depth ∞ ($n \geq 2$), inductive limits of the Cantor trees described in [8], and so on. In fact, only a countable number of countable SSTs fail the hypothesis of Theorem 2.2. One example is the SST with downward degree sequence $(1, 1, 1, \dots)$. This countable SST clearly has generalized roundness two. We do not know if the converse of Theorem 2.2 is true.

Figure 1: Theorem 2.2 and the converse question on page 5 of the source paper.

3 The Riesz–Hankel estimate

Fix $0 < \alpha < 1$. For bounded compactly supported real functions on $\mathbb{R}_+ = (0, \infty)$, set

$$\begin{aligned} A_\alpha(f, g) &= \int_0^\infty \int_0^\infty f(s)g(t)|s - t|^{-\alpha} ds dt, \\ H_{\alpha, c}(f, g) &= \int_0^\infty \int_0^\infty f(s)g(t)(s + t + c)^{-\alpha} ds dt \quad (c \geq 0). \end{aligned}$$

Write $A_\alpha(f) = A_\alpha(f, f)$.

Lemma 1 (Sharp half-line comparison). *For all such f, g and every $c \geq 0$,*

$$|H_{\alpha, c}(f, g)| \leq r_\alpha A_\alpha(f)^{1/2} A_\alpha(g)^{1/2}, \quad r_\alpha = \cos \frac{\pi\alpha}{2}.$$

In particular, A_α is a positive definite quadratic form.

Proof. We first take $c = 0$. Make the logarithmic substitution $s = e^x$ and define

$$(Uf)(x) = e^{(1-\alpha/2)x} f(e^x).$$

Direct cancellation of the powers of e^x and e^y gives

$$\begin{aligned} A_\alpha(f, g) &= \int_{\mathbb{R}^2} (Uf)(x)(Ug)(y)a_\alpha(x-y) dx dy, \\ H_{\alpha,0}(f, g) &= \int_{\mathbb{R}^2} (Uf)(x)(Ug)(y)b_\alpha(x-y) dx dy, \end{aligned}$$

where

$$a_\alpha(u) = (2|\sinh(u/2)|)^{-\alpha}, \quad b_\alpha(u) = (2\cosh(u/2))^{-\alpha}.$$

Both kernels are integrable because $0 < \alpha < 1$. With any consistent Fourier-transform convention, beta integration gives, for $z = \alpha/2 + i\xi$,

$$\begin{aligned} \widehat{b}_\alpha(\xi) &= \frac{|\Gamma(z)|^2}{\Gamma(\alpha)}, \\ \widehat{a}_\alpha(\xi) &= \Gamma(1-\alpha) \left(\frac{\Gamma(z)}{\Gamma(1-\bar{z})} + \frac{\Gamma(\bar{z})}{\Gamma(1-z)} \right). \end{aligned}$$

The reflection identity for the gamma function yields

$$\widehat{a}_\alpha(\xi) = \frac{2\Gamma(1-\alpha)|\Gamma(z)|^2 \sin(\pi\alpha/2) \cosh(\pi\xi)}{\pi} > 0.$$

Using $\Gamma(\alpha)\Gamma(1-\alpha) = \pi/\sin(\pi\alpha)$, we therefore obtain the exact quotient

$$\frac{\widehat{b}_\alpha(\xi)}{\widehat{a}_\alpha(\xi)} = \frac{\cos(\pi\alpha/2)}{\cosh(\pi\xi)} \leq r_\alpha. \quad (1)$$

Plancherel and weighted Cauchy–Schwarz now prove the asserted bilinear estimate for $c = 0$, while positivity of $\widehat{a}_\alpha$ proves positivity of A_α .

For $c > 0$, translate both zero-extended functions to the right by $c/2$:

$$f_c(u) = f(u - c/2)\mathbf{1}_{(c/2, \infty)}(u), \quad g_c(u) = g(u - c/2)\mathbf{1}_{(c/2, \infty)}(u).$$

Then $H_{\alpha,c}(f, g) = H_{\alpha,0}(f_c, g_c)$, whereas translation invariance of $|s-t|^{-\alpha}$ gives $A_\alpha(f_c) = A_\alpha(f)$ and similarly for g . The result follows. $\square$

The same estimate applies when either function is supported on a finite interval: extend it by zero to the half-line. Reversing an interval coordinate does not change its diagonal Riesz energy.

4 Flows on a finite-topological tree

Let G be a finite-topological tree with open edges $e = 1, \dots, N$. Choose a root and orient each edge away from it. For a finitely supported signed measure

$$\mu = \sum_{k=1}^m a_k \delta_{x_k}, \quad \sum_{k=1}^m a_k = 0,$$

define $F_e(s)$ to be the μ -mass of the component lying beyond the point at coordinate s on edge e . On every half-line F_e has compact support. At a branch vertex, the incoming value minus the sum

of the outgoing values is exactly the atom of μ at that vertex. Consequently, edgewise integration by parts gives

$$\int_G \varphi d\mu = \sum_{e=1}^N \int_e F_e(s) \partial_e \varphi(s) ds \quad (2)$$

for every continuous function that is continuously differentiable on open edges. This identity incorporates atoms at vertices through the preceding Kirchhoff balance; there are no omitted boundary terms.

Lemma 2 (Finite-edge coercivity). *Let $1 < p < 2$, put $\alpha = 2 - p$, and let $r_p = \sin(\pi(p-1)/2) = \cos(\pi\alpha/2)$. Then*

$$-\int_G \int_G d(x,y)^p d\mu(x) d\mu(y) \geq p(p-1)(1 - (N-1)r_p) \sum_{e=1}^N A_\alpha(F_e). \quad (3)$$

In particular, if $(N-1)r_p \leq 1$, then G has p -negative type.

Proof. Apply (2) in both variables. Since $\alpha \in (0,1)$, all resulting second-derivative kernels are locally integrable, so the calculation is justified by ordinary approximation or Fubini's theorem. On a single edge,

$$-\partial_s \partial_t |s-t|^p = p(p-1)|s-t|^{-\alpha}.$$

For two distinct open edges $e \neq f$, the path joining their interior points exits each edge through a fixed endpoint. If u, v denote distances from those endpoints, then

$$d(x_e(u), x_f(v)) = u + v + c_{ef}$$

for some $c_{ef} \geq 0$. Relative to the globally chosen orientations, u or v may be reversed; hence the corresponding mixed derivative has an inessential sign. It follows that

$$\begin{aligned} -\int_G \int_G d(x,y)^p d\mu(x) d\mu(y) &= p(p-1) \left(\sum_e A_\alpha(F_e) \right. \\ &\quad \left. + \sum_{e \neq f} \varepsilon_{ef} H_{\alpha, c_{ef}}(\tilde{F}_e, \tilde{F}_f) \right), \end{aligned}$$

where $\varepsilon_{ef} \in \{-1, 1\}$ and each tilde records at most a coordinate reversal and zero extension.

Lemma 1 bounds the absolute value of the (e, f) term by $r_p A_\alpha(F_e)^{1/2} A_\alpha(F_f)^{1/2}$. Therefore

$$\begin{aligned} \frac{-\int \int d(x,y)^p d\mu(x) d\mu(y)}{p(p-1)} &\geq \sum_e A_\alpha(F_e) - r_p \sum_{e \neq f} \sqrt{A_\alpha(F_e) A_\alpha(F_f)} \\ &\geq (1 - (N-1)r_p) \sum_e A_\alpha(F_e). \end{aligned}$$

The last inequality follows from $(\sum_e \sqrt{A_e})^2 \leq N \sum_e A_e$. This proves (3). When its coefficient is nonnegative, the defining zero-sum quadratic inequality for p -negative type follows. $\square$

Theorem 3 (Finite-topological trees have roundness above one). *Every finite-topological tree has generalized roundness strictly greater than one. More explicitly, if it is presented with N open edges, then it has p -negative type for every $1 < p < 2$ satisfying*

$$(N-1) \sin \frac{\pi(p-1)}{2} \leq 1. \quad (4)$$

For example, the universally safe choice

$$p = 1 + \frac{2}{\pi} \arcsin\left(\frac{1}{2 \max\{1, N-1\}}\right)$$

is strictly larger than one.

Proof. Lemma 2 proves the zero-sum quadratic inequality for every finite configuration in G . This is p -negative type. Generalized roundness equals supremal negative type, as recalled in the source paper [1], so the displayed safe exponent gives $\gamma(G) > 1$. $\square$

5 Resolution of the converse

Corollary 4 (Converse of source Theorem 2.2). *Let (T, ρ) be a countable spherically symmetric tree with downward degree sequence $(d_0, d_1, \dots)$. Then*

$$\gamma(T, \rho) = 1 \iff |\{j : d_j > 1\}| = \aleph_0.$$

Proof. The reverse implication is Theorem 2.2 of [1]. Conversely, suppose that only finitely many d_j exceed one. Because the tree is countably infinite, its downward degrees are positive; after the last branching level every descendant chain is a ray. There are finitely many such rays and only a finite subtree before them. Suppressing degree-two vertices in its geometric realization therefore presents it as a finite-topological tree with some finite number N of open edges. The original vertex set T is an isometric subset of that realization. By Theorem 3, the realization, and hence its subset T , has p -negative type for some $p > 1$. Thus $\gamma(T) > 1$. $\square$

6 Verification and limitations

The proof gives a lower bound depending on the chosen finite edge decomposition, not the exact generalized roundness. Subdividing edges makes the bound weaker but cannot invalidate it. The argument also proves the stronger result for arbitrary finite-topological trees, without spherical symmetry or unit edge lengths.

The accompanying script `code/verify_finite_edge_bound.py` checks the closed multiplier ratio on five α values and 4,001 frequencies per value, and independently compares the beta/gamma formulas with that closed ratio at 25 parameter pairs. It also tests the explicit safe exponent on 2,400 random weighted finite trees by compressing the powered distance matrix to the zero-sum subspace. These checks are designed to catch sign, factor-of-two, and edge-count errors; they are not used in the proof.

The points most deserving human verification are (i) the edgewise integration-by-parts identity at branch vertices, and (ii) the fact that the sum over distinct edges is an ordered sum, matching the factor $N - 1$ in the final Cauchy–Schwarz estimate.

7 Bounded novelty check

On 27 August 2026, the run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:1008.3418, its title, the exact converse wording, and finite-ended/finitely-branching negative-type variants. Bounded arXiv and web searches used the same phrases and inspected the related arXiv papers 1608.03699, 0901.0695, and 1108.0451. The searches found the original question and

adjacent results on generalized roundness and finite metric spaces, but no exact answer to this converse and no matching finite-topological-tree theorem. This makes novelty plausible but does not certify it.

References

- [1] E. Caffarelli, I. Doust, and A. Weston, *Metric trees of generalized roundness one*, Aequationes Math. 83 (2012), arXiv:1008.3418v2, doi:10.1007/s00010-011-0108-8.